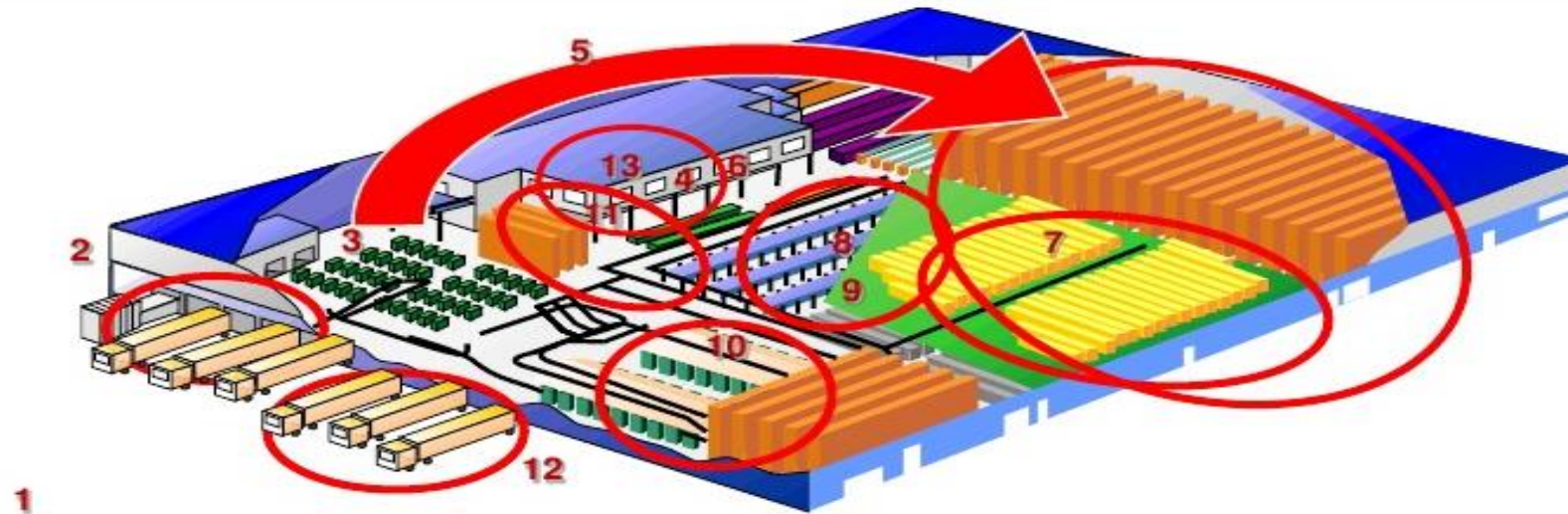


CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE



- 1 - Inbound
- 2 - Dock / Unload
- 3 - Receipt of goods
- 4 - Create GRN
- 5 - Putaway
- 6 - Picklist Creation

- 7 - Order picking
- 8 - Value-added services (Kitting)
- 9 - Rework
- 10 - Packing line
- 11 - Dispatch
- 12 - Tracking
- 13 - MIS & Documentation

Development of the Common Standard Curricula on International Transport and Logistics Basic Training for ASEAN Member States under Sustainable Human Resource Development in Logistics Services

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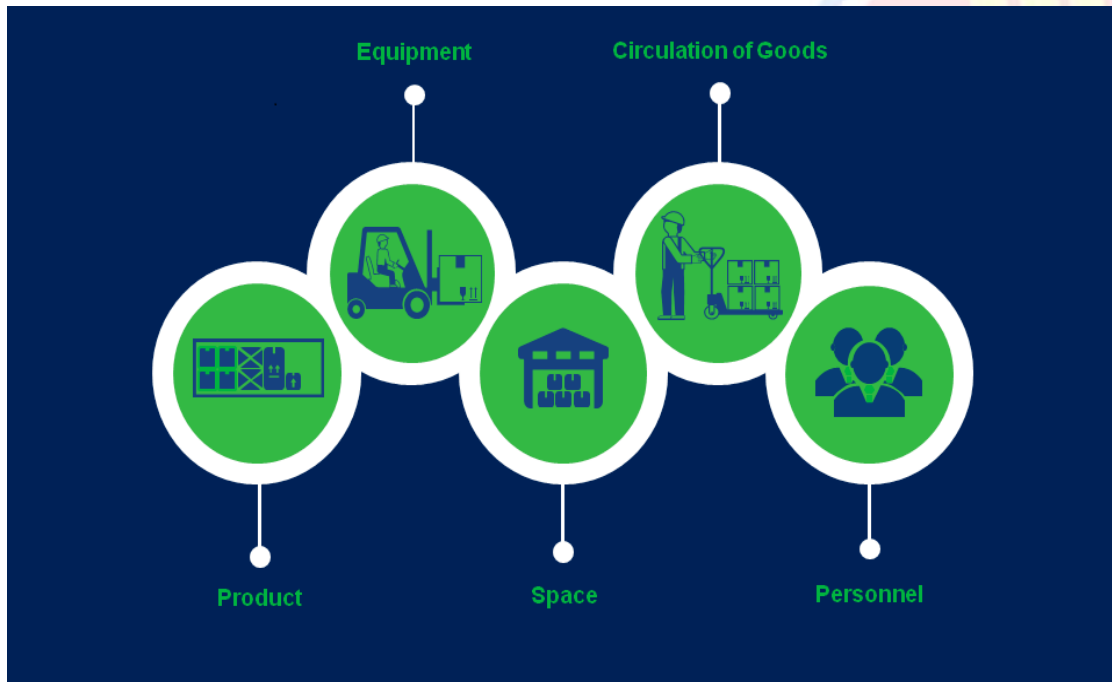
CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

MODULE CONTENT

- Types of warehouse layout
- Factors determining warehouse layout
- Storage systems used in the warehouse
- Types of MHE and their specifications

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

FACTORS AFFECTING WAREHOUSE DESIGN

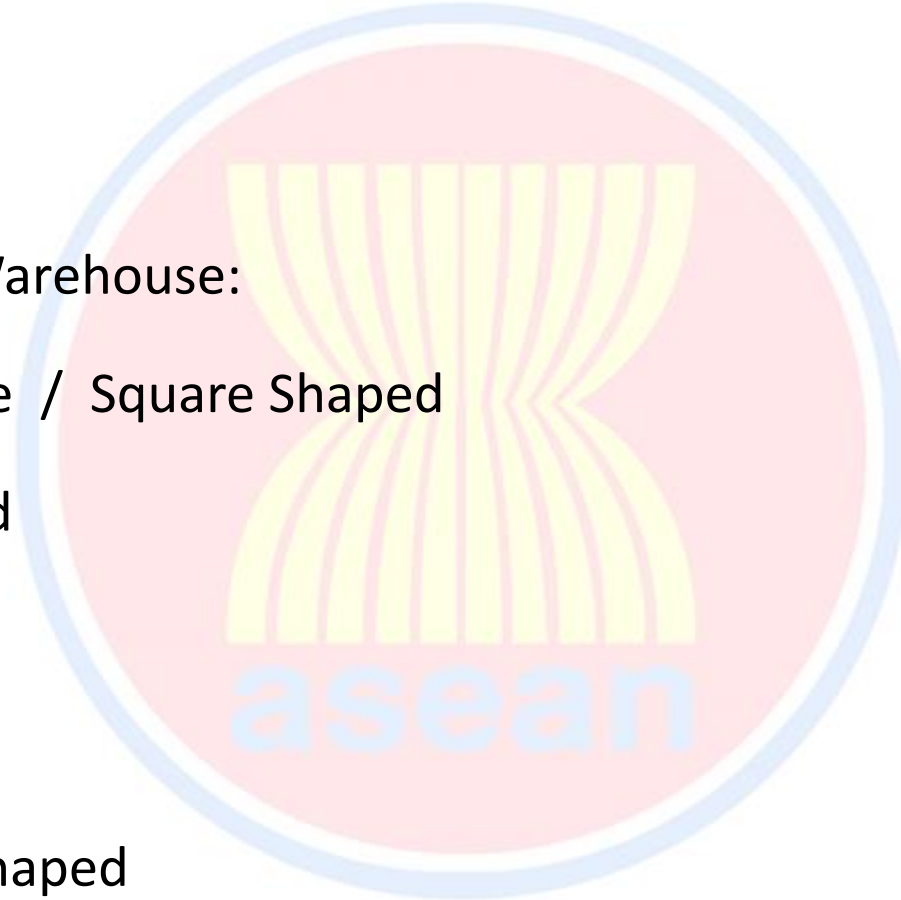


- 1.**Product:** What kind of goods will it hold?
- 2.**Space:** What physical characteristics should the warehouse have?
- 3.**Equipment:** What MHEs are available?
- 4.**Circulation of goods:** How do goods circulate?
- 5.**Personnel:** How many employees are there?

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

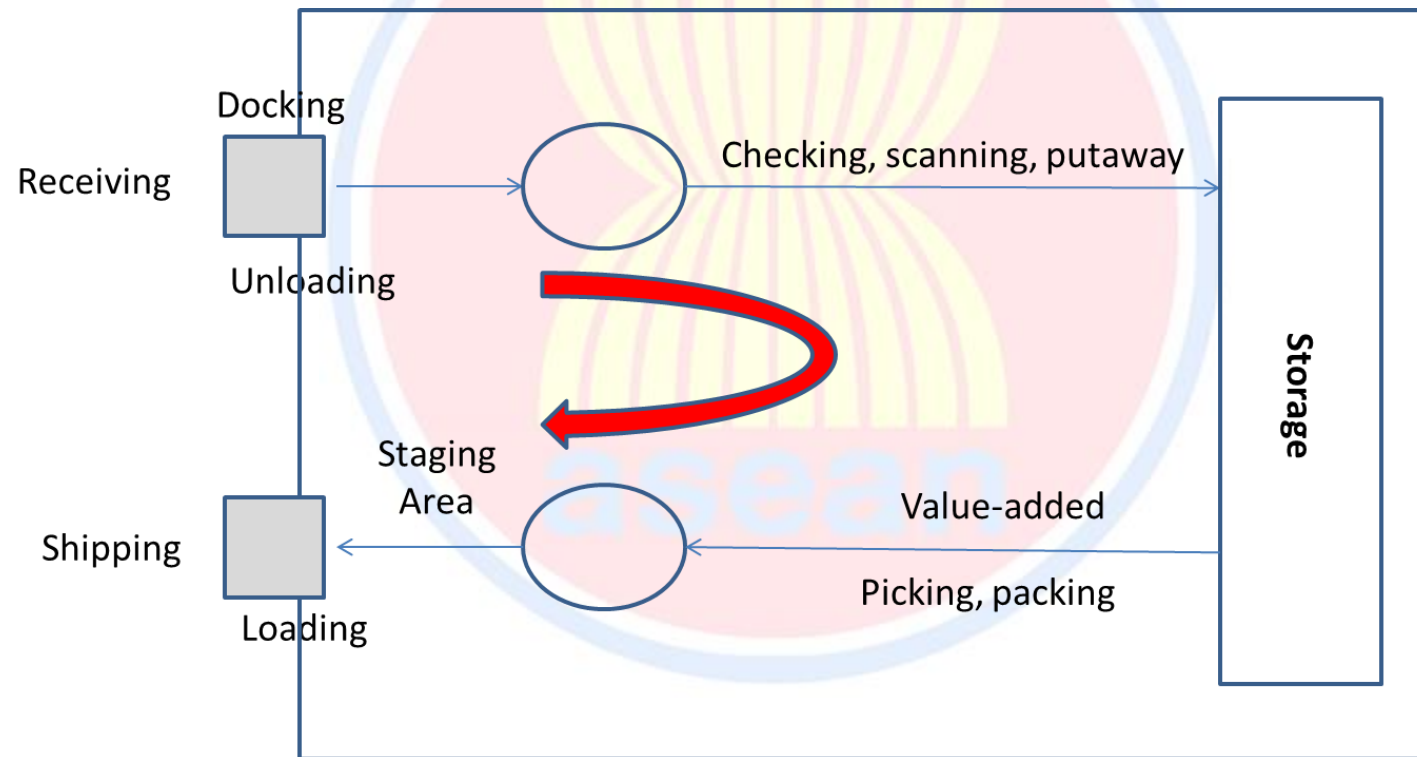
ALLOCATION OF SPACE

- Shapes of Warehouse:
 - Rectangle / Square Shaped
 - U-Shaped
 - I-Shaped
 - L-Shaped
 - Round-Shaped



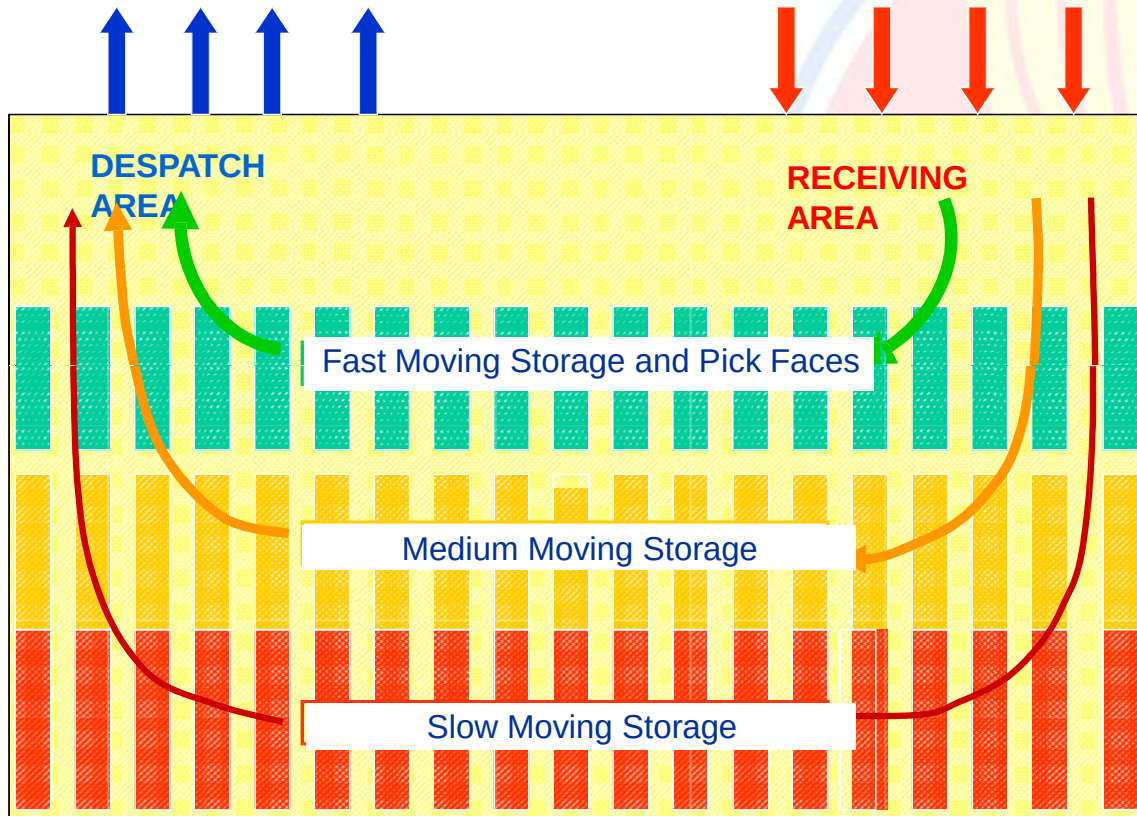
CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

WAREHOUSE LAYOUT – U SHAPE FLOW



CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

WAREHOUSE LAYOUT – U SHAPE FLOW (CON'D)



- U-shape flow occurs when the goods receipt and dispatch functions are located at the same side/end of a warehouse building.
- Products flow in at receiving, move in to storage in the back of the warehouse, and then to shipping, which is located at the adjacent to receiving on the same side of the building.
- Items with higher throughput level are located closer to the loading bays.

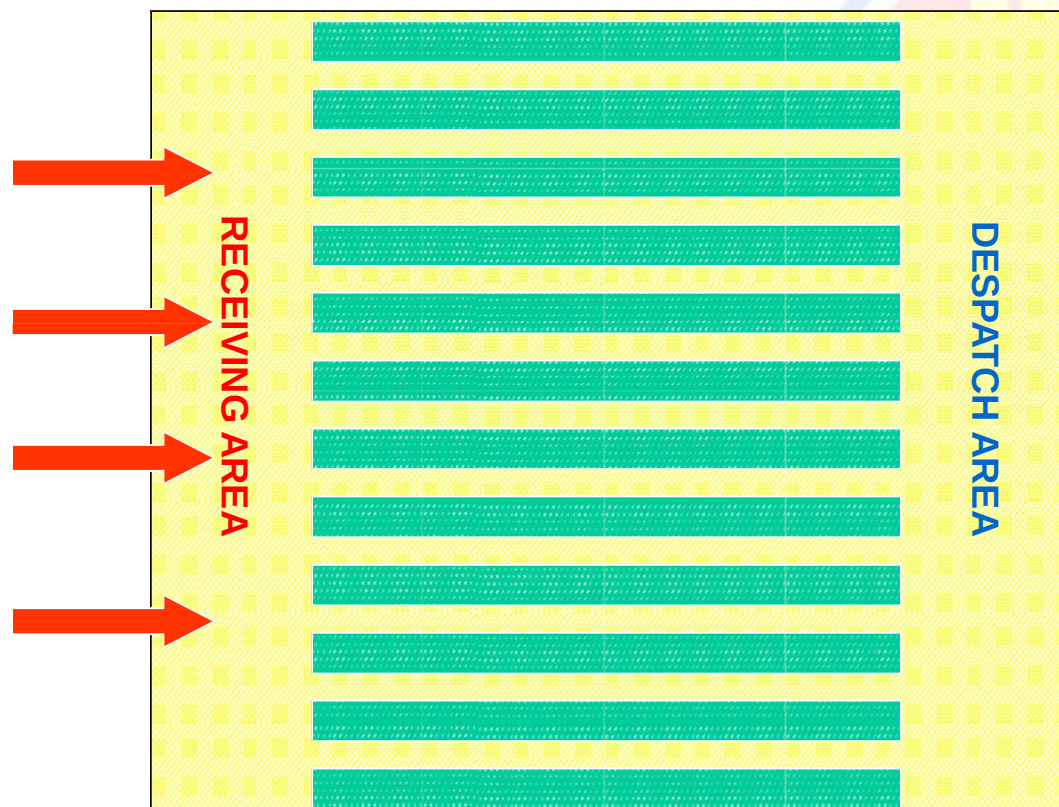
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ADVANTAGES OF U SHAPE FLOW

- Excellent utilization of dock resources because the receiving and shipping processes can share dock doors
- Facilitating cross-docking because the receiving and shipping docks are adjacent to one another and may be co-mingled
- Excellent lift truck utilization because put away and retrieval trips are easily combined and because the storage locations closest to the receiving and shipping docks
- High degree of security control

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

WAREHOUSE LAYOUT – I SHAPE FLOW (THROUGH FLOW)



- 'Through' flow happens when separate loading bay facilities for outbound and shipping are provided, often at opposite end of warehouse.
- Products flow in at receiving, move into storage, picking area and then the marshalling and despatch area in a straight line.
- Items with a higher throughput level are located at the center of the warehouse.

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

APPLICATION OF I SHAPE FLOW

- When there is a risk of interference or confusion between goods in and goods out.
- When goods inwards vehicles and dispatch vehicles are very different; for example differences in platform height or nature of unit load.
- When a warehouse is connected to a production plan

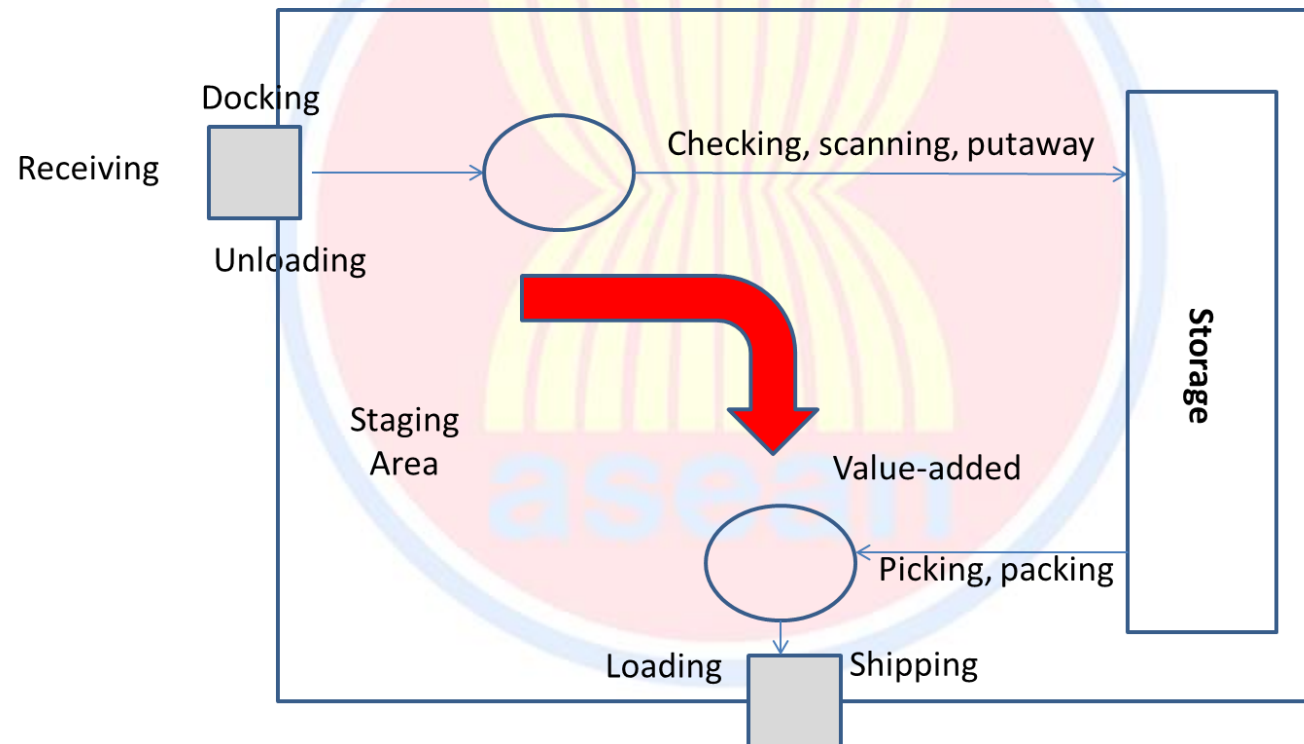
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DISADVANTAGES OF I SHAPE FLOW

- The major disadvantage of a 'Through' flow layout is goods need to travel the full length of the warehouse, even for goods that have a higher throughput level.
- It is also harder to control and less flexible.

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

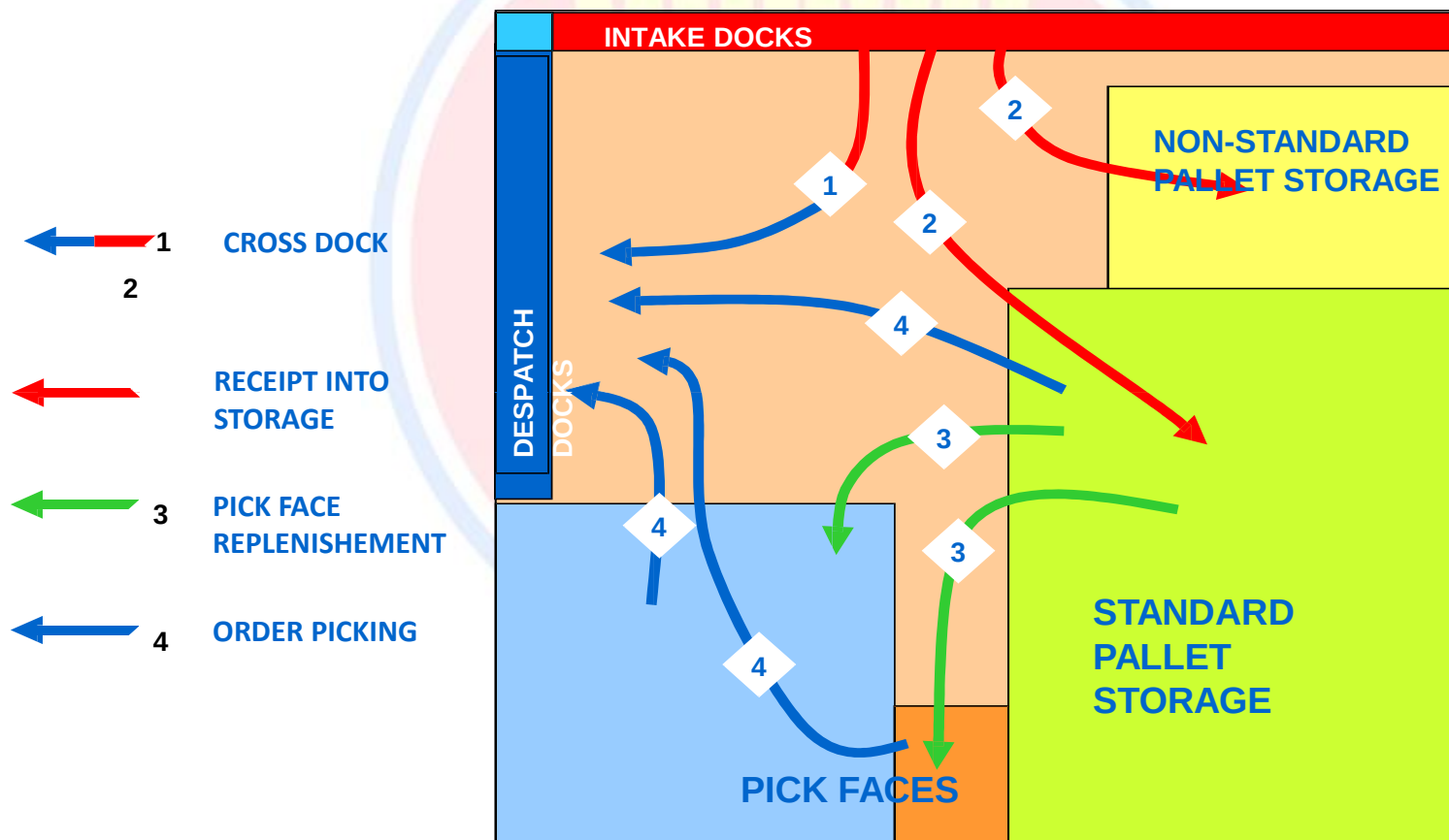
WAREHOUSE LAYOUT – L SHAPE FLOW



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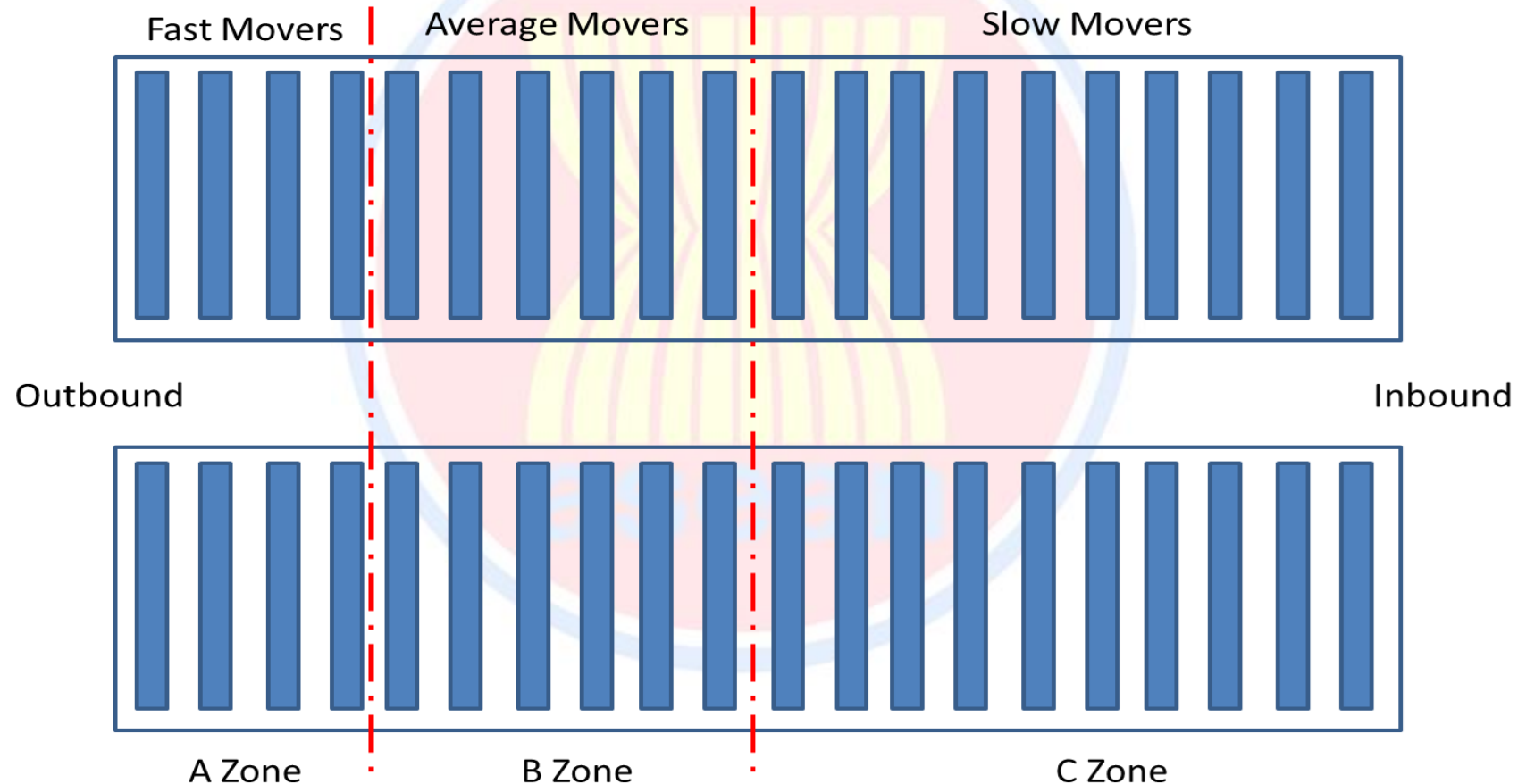
WAREHOUSE LAYOUT – COMPLEX FLOW

Mixed Storage and Cross-dock requirement



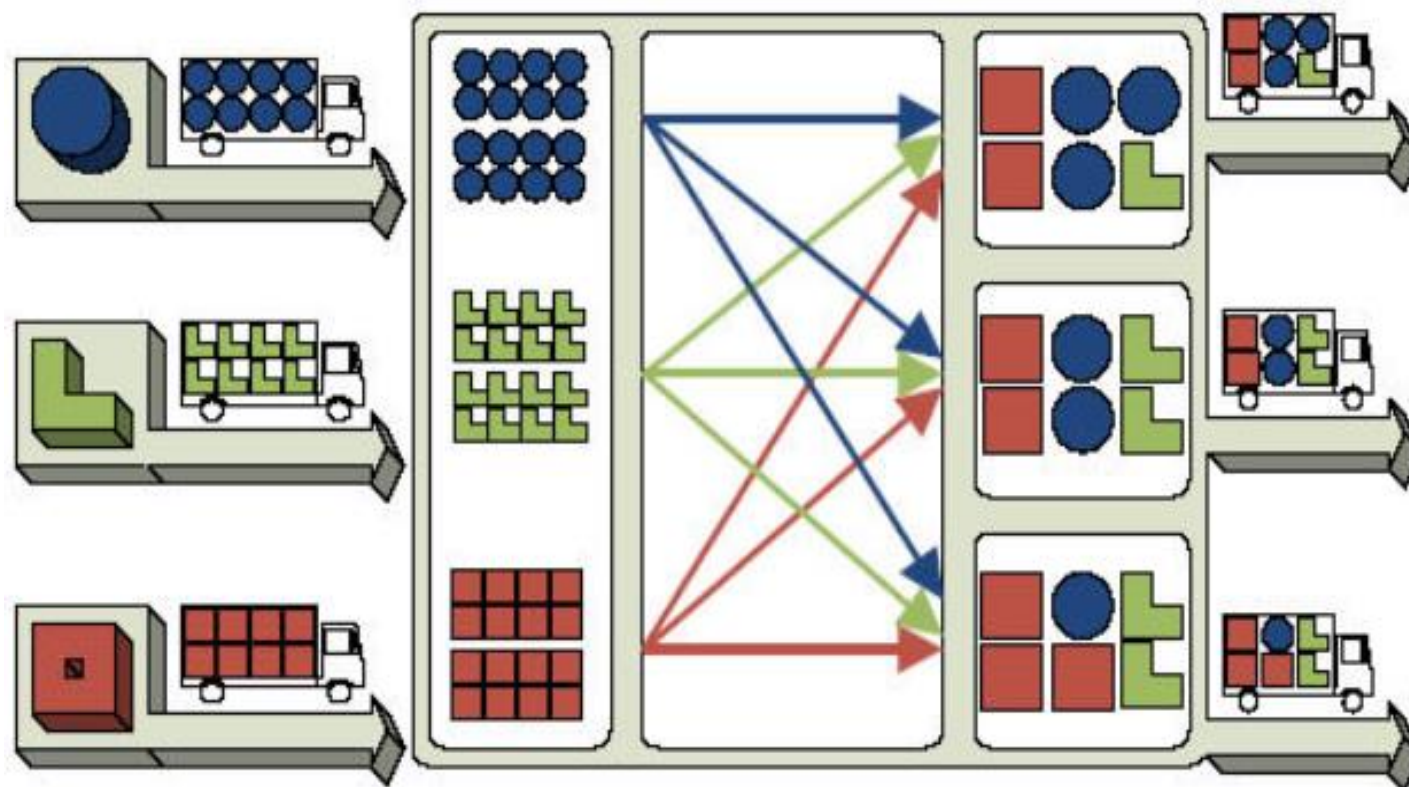
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WAREHOUSE LAYOUT – ABC LAYOUT



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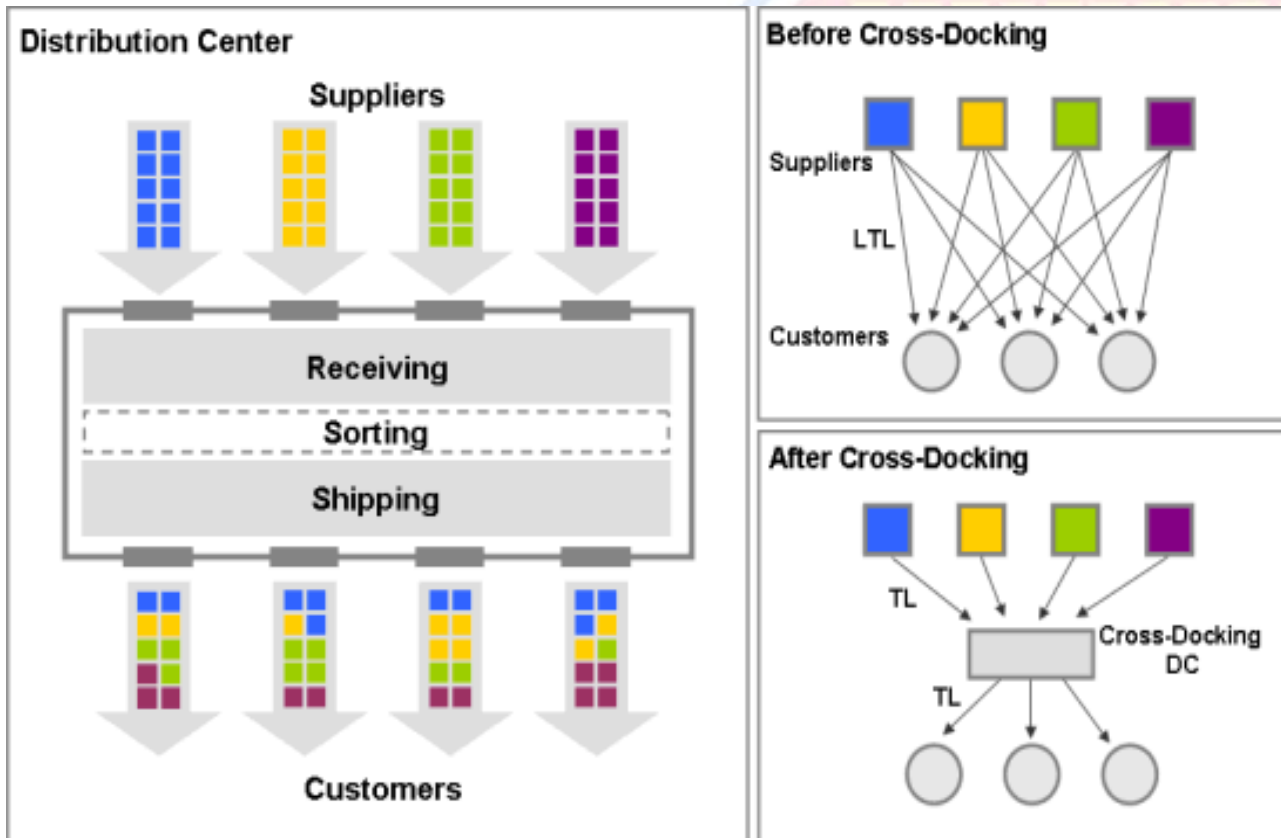
WAREHOUSE LAYOUT – CROSS-DOCK



Cross Docking Warehouse

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WAREHOUSE LAYOUT – CROSS-DOCK (CON'D)



Cross-docking is used when:

- Inventory demand is stable and consistent
- Time sensitive and perishable inventory
- Variable product purchase

WATCH CLIP:

<https://www.youtube.com/watch?v=ubGEt9V09bo>

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BENEFITS OF CROSS-DOCKING

- Minimize warehousing space
- Reduce inventory carrying cost
- Reduce labor cost
- Reduce damages due to minimal handling
- Shorten delivery time
- Increase customer service level
- Reduce transportation cost
- Reduce fixed asset costs

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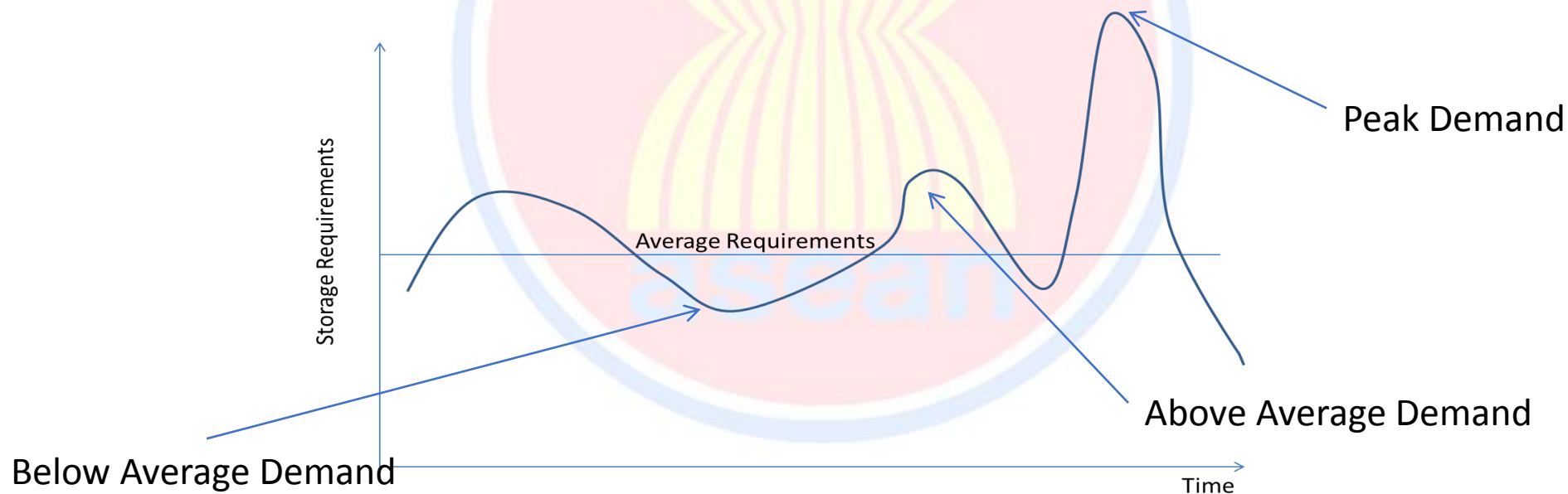
TYPES OF CROSS-DOCKING

- **Manufacturing cross-docking:**
Receiving of materials for production
- **Distributor cross-docking:**
Consolidate inbound products from different vendors and deliver to distributors
- **Transportation cross-docking:**
Combine shipments from a number of carriers or LTL shipment of small loads to gain economies of scale
- **Retail cross-docking:**
Receiving of products from multiple vendors and sorting into outbound trucks for retail stores.

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WAREHOUSE LAYOUT – SPACE REQUIREMENT

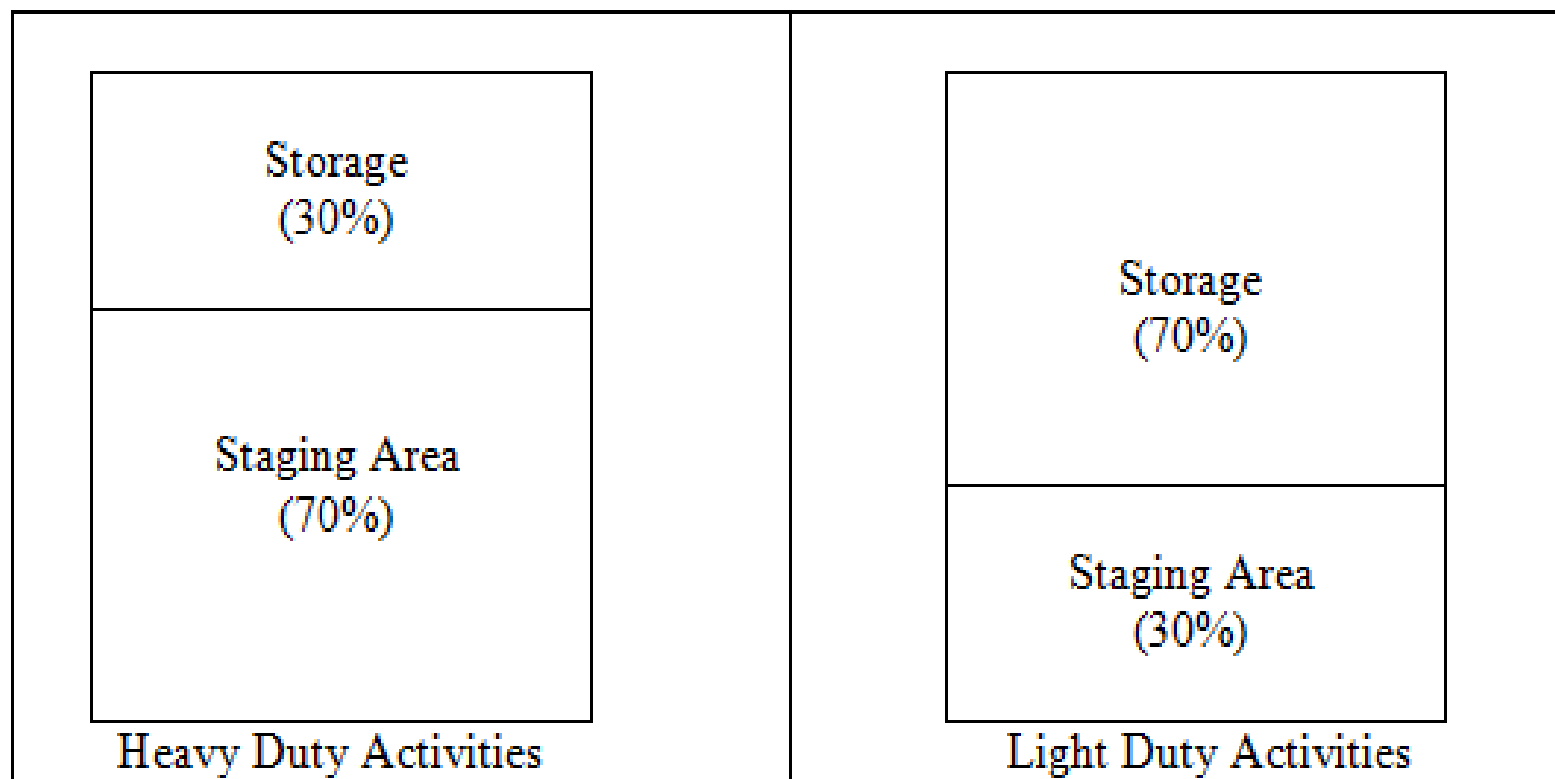
Part	Jan-19	Feb-19	Mar-19	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19
Average Inventory (Cbm)	4,000	8,000	5,000	3,000	4,000	5,000	9,000	5,000	4,500	8,000	10,000	3,500



SEASONAL FACTORS THROUGHOUT THE YEAR

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

WAREHOUSE LAYOUT – SPACE REQUIREMENT (CON'D)



CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

AREAS WITHIN THE WAREHOUSE

Sq	Area	Floor Space (sqm)	Remarks
1	Dock Leveler area		
2	Receiving area		
3	Storage area + aisle		
4	Staging area		
5	Packing area		
6	Shipping area		
7	Emty pallet storage area		
8	MHE charging area		
9	Office		
10	Rest room		

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

LAYOUT DESIGN PRINCIPLES CONSIDERATIONS

- There are 4 factors must be considered when designing a layout:
 - Customer service level.
 - Material flow.
 - Low cost and reasonable.
 - Types of activities.

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

LAYOUT DESIGN PRINCIPLES CONSIDERATIONS (CON'D)

- The most commonly accepted principles of warehouse design and layout are as follows:
 - Use a one-store facility.
 - Use straight-line or direct flow.
 - Use efficiently materials handling equipment and operations.
 - Use storage plan effectively.
 - Keep aisle space to a minimum.
 - Utilize maximum the height of the building.
 - All materials should be easily observable.

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

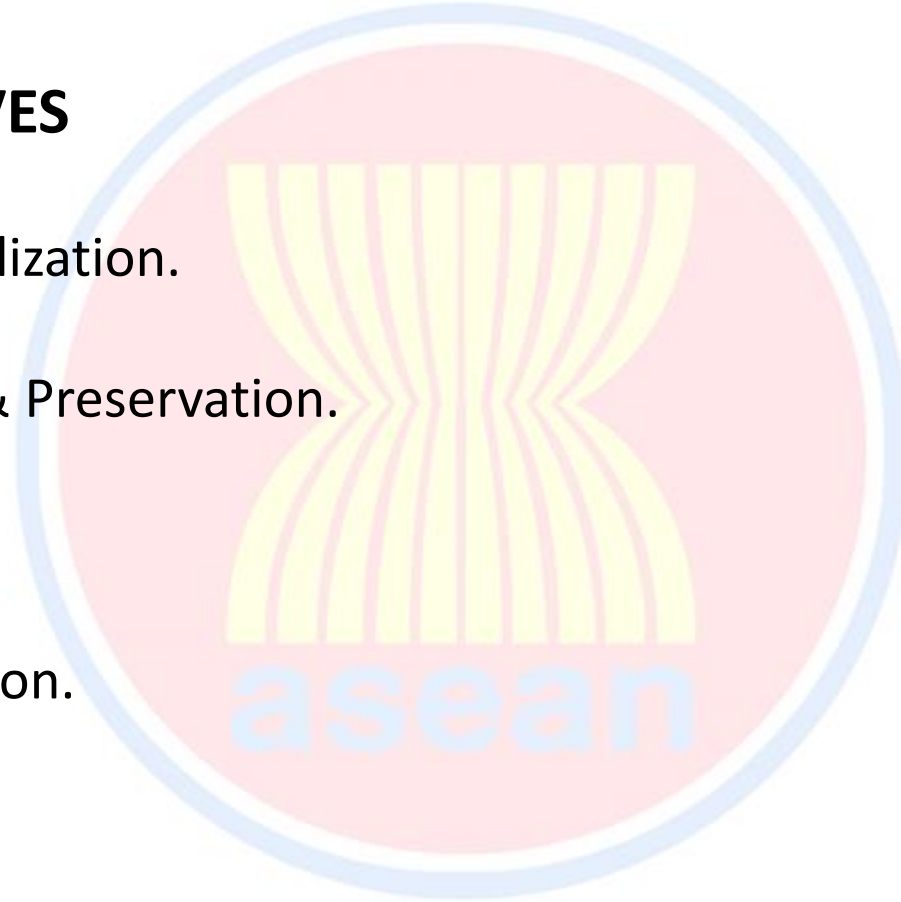
LAYOUT DESIGN PRINCIPLES CONSIDERATIONS (CON'D)

- Accessibility.
- All movements should be both necessary and direct.
- Work lanes and transport lanes must not cross each others.

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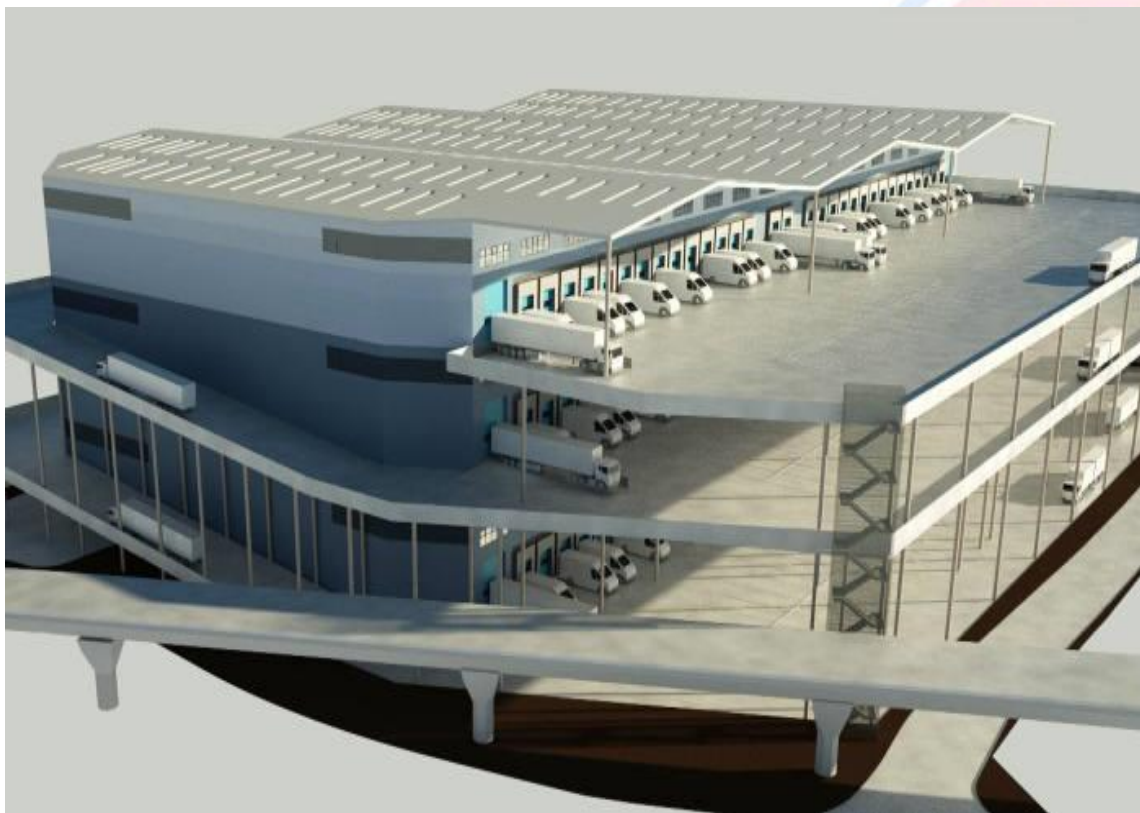
LAYOUT DESIGN OBJECTIVES

- Capacity Utilization.
- Protection & Preservation.
- Efficiency.
- Mechanization.
- Resources.



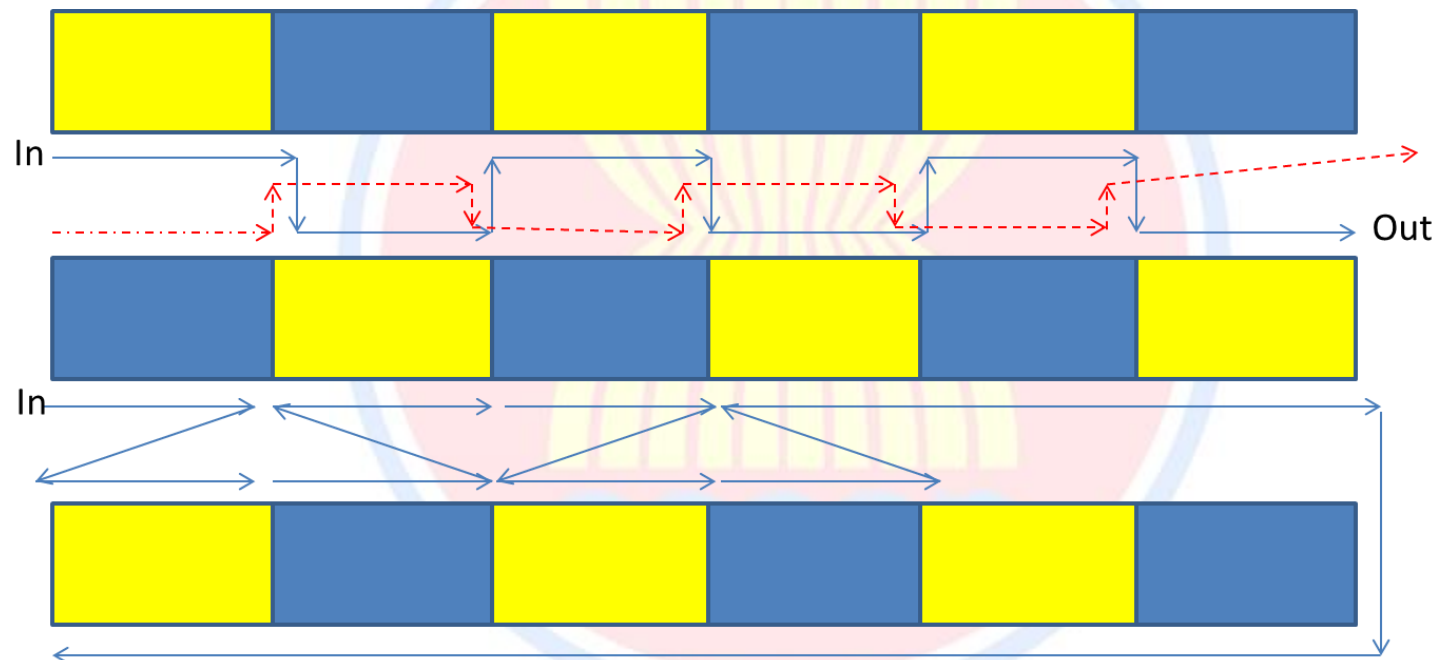
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WAREHOUSE LAYOUT – MULTI STORIES



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PICK ROUTING



Even Odd Picking

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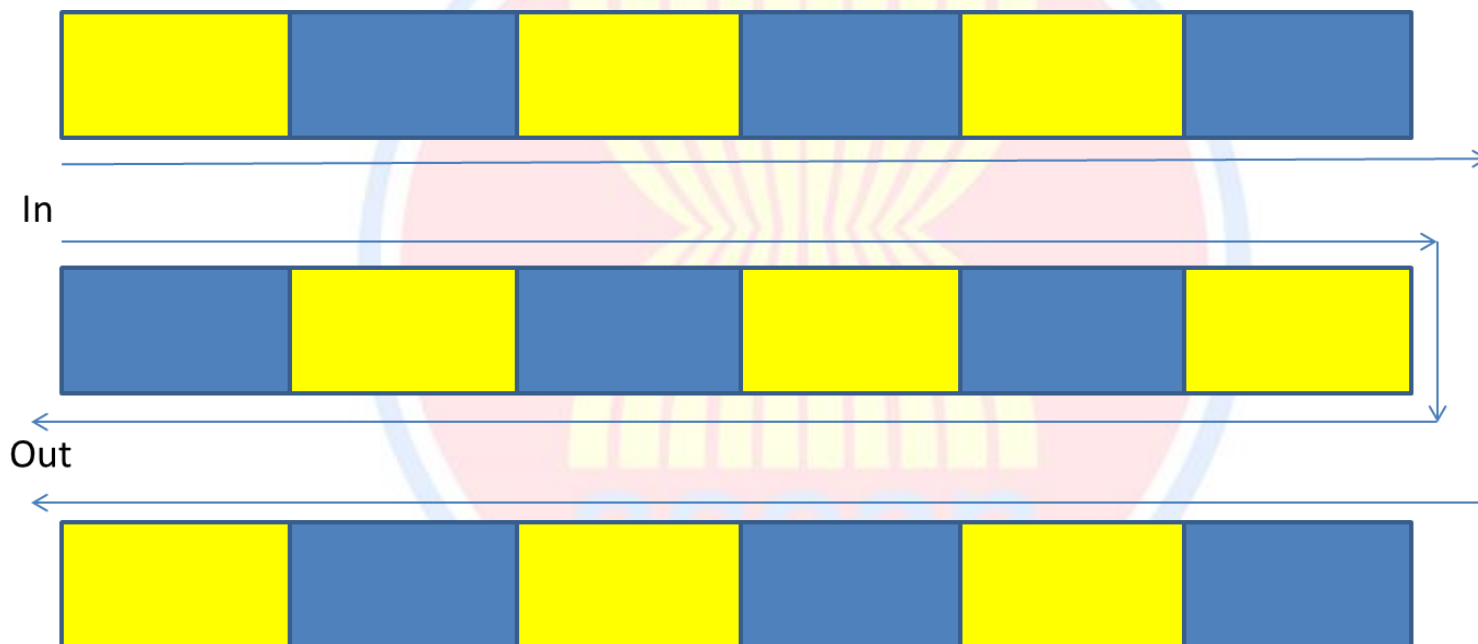
PICK ROUTING (CON'D)



Single Pick Path Single Order Picker

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PICK ROUTING (CON'D)



Single Pick Path Dual Order Pickers

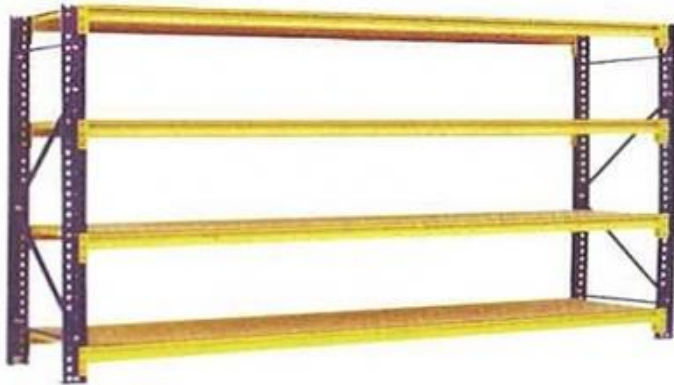
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STORAGE & HANDLING

GROUP	TYPES OF RACK
GROUP 1: Small items, issued in detail with little or no MHE involvement e.g. Pharmaceuticals, Cosmetics	Shelving (bin, drawers, plastic boxes)
GROUP 2: Larger items or Packages, MHE Required with receipt storage and issue possibly by unit load	Light racking, long span shelving Pallet racking
GROUP 3: Large Items, handling possible only with MHE, special equipment or attachments may be required. Generally presenting a storage and handling problem	Pallet, Pallet racking
GROUP 4: Items stored and handled in Bulk e.g. fuels, lubricants, grain, sugar, powders	Bulk containers

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

SHELVING STORAGE



**MULTI-
FUNCTIONAL
MODULAR
SHELVING**



- Small items pick
- Picking done by hand

**LONG
SPAN
SHELVING**



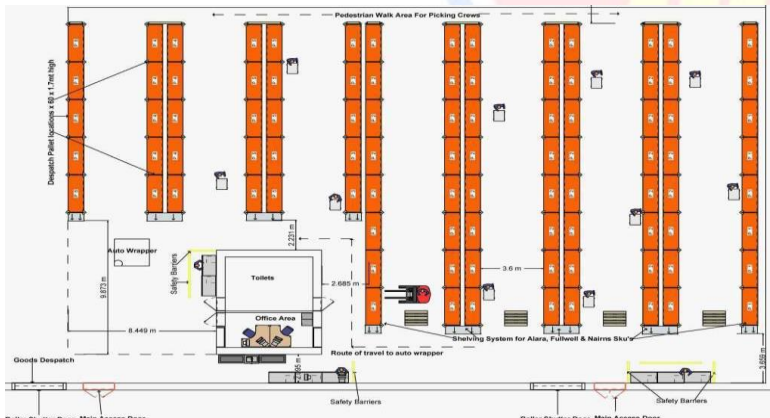
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SELECTIVE RACKING



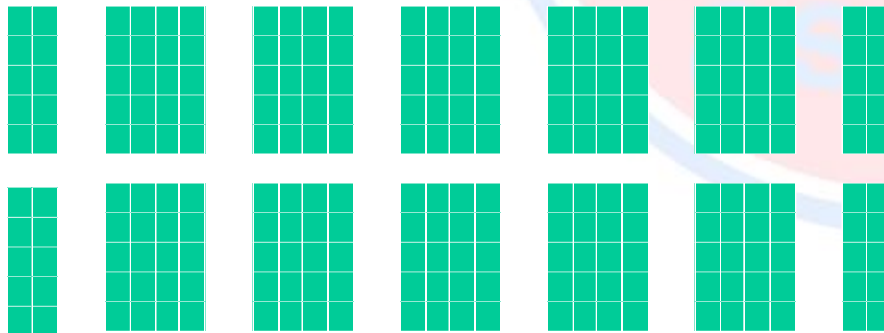
Conventional pallet racking

- Versatile storage solution
- Access inventory easily
- Support FIFO order picking
- Ease of access of different types of MHEs
- Able to accommodate all types of handling system



CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

DOUBLE DEEP RACKING

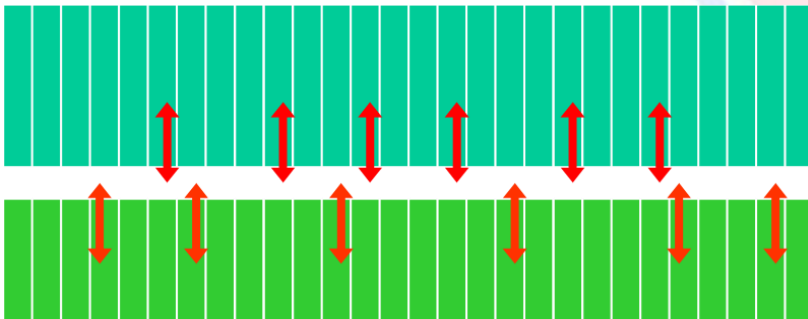


Conventional pallet racking

- Efficient use of floor space.
- Increase storage density by storing two-deep (30-40% increase); require double deep truck.
- Keep product line together by storing 2 pallets deep
- Picking done by conventional trucks or reach trucks.

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

DRIVE - IN RACKING



Drive-in racking

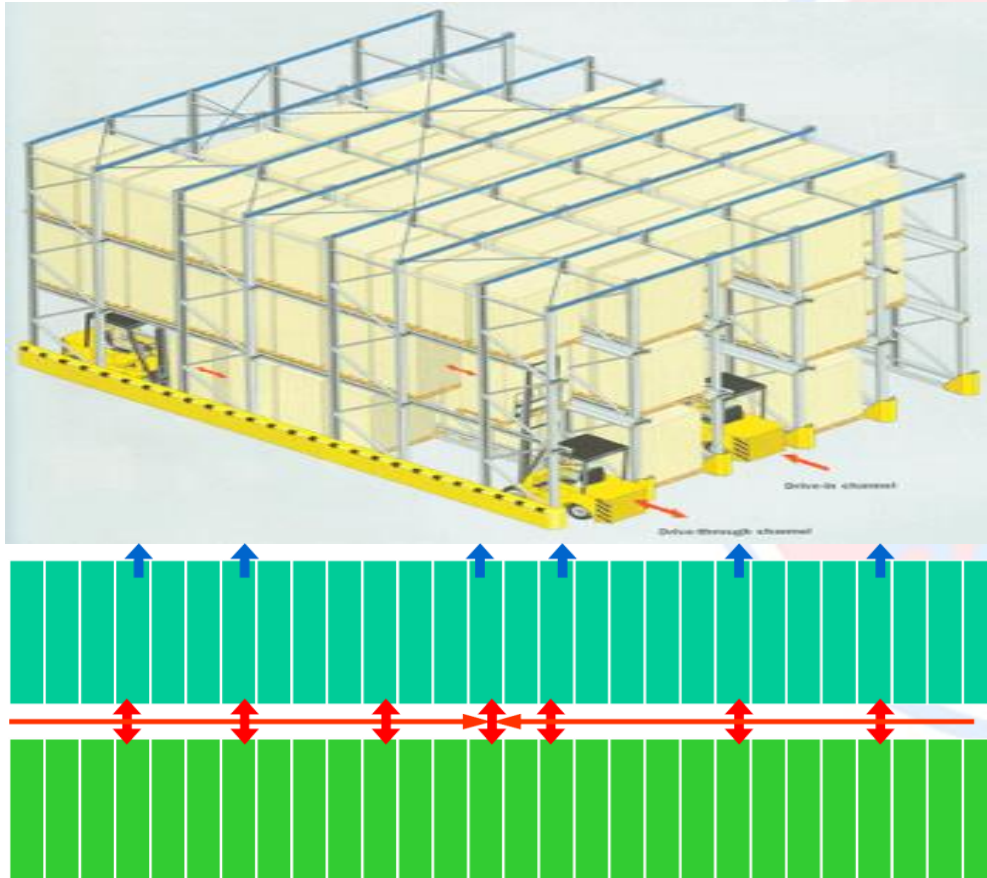
- Good storage density can be achieved.
- Loading and unloading from the same side of the rack
- Good for pallet loads placed against one end of wall, meaning a single stowing and retrieval aisle
- Support loads on short arms for truck access.
- Cannot mix products since access is from one direction.
- Permits use of LIFO.

WATCH CLIP:

<https://www.youtube.com/watch?v=CmWX2kqPn2M>

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

DRIVE-THROUGH RACKING



Drive-through racking

- Load from one side and unload from the other side.
- Good for pallet loads with access at both ends of tunnel.
- High storage density can be achieved.
- Truck access from two directions suggesting picking and put away are independent of each other.
- Permits use of FIFO.

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CANTILEVER RACKING



Cantilever racking

- Row of single upright columns with arms extending from one or both sides.
- Suits storage of long, bulky loads, such as:
 - Lumber
 - Steel bar & pipes
 - Furniture
 - Sheet metal

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MOBILE RACKING



Mobile racking

- Manual, mechanical assist or electronic controlled.
- Reduces aisles and saves up to 50% of floor space.
- Based on floating aisle, only one needed per row.
- Used for loose materials or unitized packages.
- Allows large selection to order picker in a concentrated area.

WATCH CLIP:

<https://www.youtube.com/watch?v=OxNLQyv8wiw>

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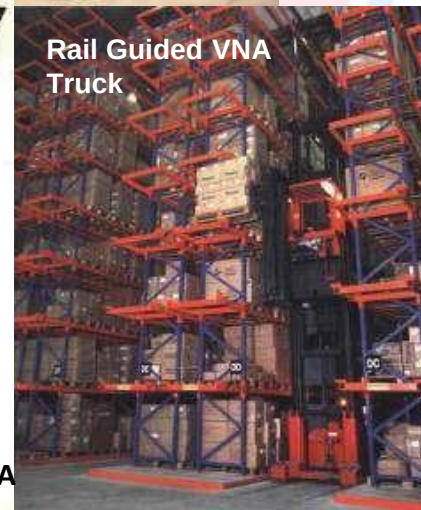
VNA RACKING (VERY NARROW AISLE)



Super Flat
Floor

Tolerance:

1mm in
1000mm



VNA racking

- Improve space utilization
- Improve MHEs productivity
- Special lift truck is required

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

AS/RS RACKING



AS/RS (Auto Storage & Retrieval System)

- Uses a fixed path system to put away and retrieve pallets from fixed rows of racks.
- In a typical cycle, the AS/RS picks a pallet at the I/O (input/output) station and the SR (storage retrieval) machine stows the pallet the pre-determined empty location by the software.
- Land and building cost savings.
- Saving of labour cost
- Better inventory accountability.
- High safety
- Increased accuracy, productivity & efficiency
- Space optimization

WATCH CLIP:

<https://www.youtube.com/watch?v=n3nQymCx7U>

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

BULK STORAGE



*BULK STORAGE IN
CARTON*



*BULK STORAGE IN
BAG*



*GRAIN BULK
STORAGE*

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

COMPARISON ANALYSIS OF RACKING TYPES

Type	Flexibility	Density	Selectivity	Capital	Remarks
Bulk	5	3	2	5	
Cantilever	4	1	5	3	
Single deep	3	1	5	4	One sided access
Pallet gravity	3	4	3	2	FIFO, 2-sided access
Double deep	2	2	2	3	1-sided access
Drive through	1	4	3	5	1 SKU per bay, 2-sided access
Push back	1	4	3	2	LIFO, 1-sided access
AS/RS	1	5	5	1	Program to increase flexibility

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

DOCKING SYSTEM



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DOCK LEVELER

- The dock leveler is an equipment that warehouse staff uses to level a container and the dock when loading or unloading.
- The function of the dock leveler is to move up or down, depending on where the trailer bed is located so machines such as forklifts can travel across to the dock smoothly.



CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

COMPUTING NUMBER OF DOCKS

Example of calculating number of docks / doors:

- Working hours per day: 8hrs
- Loading (unloading rate): 1.6 hrs / TEU
- Number of TEUs per dock per day: $8 : 1.6 = 5$ TEUs
- Number of containers (TEU) per day: 20 TEUs

NUMBER OF DOCKS / DOORS REQUIRED: $20 : 5 = 4$ docks / doors

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COMPARISON OF STORAGE MODEL

BLOCK STACKING	◆ Cheap to set up ◆ Uses simple equipment ◆ Makes good use of cube in low level buildings	▪ Risk of damage to product ▪ Control problems ▪ FIFO Difficult ▪ Poor use of cube in high buildings
DRIVE-IN / DRIVE-THROUGH	◆ Permits high stacking of fragile product. ◆ Uses simple MHE (FLT's) ◆ Good use of cube	▪ Needs strong pallets ▪ Slow in operation ▪ Needs good floors ▪ High risk of damage by trucks ▪ Expert design and construction ▪ Can have stock rotation problems (DI) ▪ Limited pallet accessibility
DOUBLE DEEP	◆ Good cube utilization	▪ Special trucks required ▪ Tight control to achieve FIFO ▪ Accessibility to only 50% of locations

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

COMPARISON OF STORAGE MODEL (CON'D)

VNA	◆ Excellent use of cube ◆ High pallet accessibility. ◆ Small area footprint; good floor space utilization	▪ Equipment expensive ▪ Usually requires P&D stations and supporting FLTs ▪ Needs very high quality floor ▪ Usually requires rail of wire guidance ▪ Expert design and construction
MOBILE RACKING	◆ Extremely good use of cube ◆ Good security of product	▪ High cost ▪ Equipment cost high ▪ Must be designed and installed by experts. ▪ Equipment needs maintenance ▪ Slow operation, work flow must be balanced

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

MATERIAL HANDLING EQUIPMENT (MHE)



Reach
Truck



Very Narrow
Aisle (VNA)
Stacker Truck



Pedestrian
Operated
Powered
Stacker Truck
(POPST)



Hand Pallet
Truck or
"Pallet Jack"



VNA High-
Rise
Stacker
Crane



Counterbalance
Truck

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

HAND PALLET TRUCK

Hand Pallet Truck	Characteristics
	• Useful where mechanized MHE cannot be used. • Employed where space limitations persist, aisle width, 2-4ft. • Good for movement of simple items.

Hand Pallet Truck

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

POWER PALLET TRUCK

Powered Pallet Truck



Characteristics

- Used to transport pallets of heavy items.
- Intermittent duty and wide range of applications.
- Good in highly congested areas, where maneuverability is important.
- Capacities range to 5t.
- Travel distance up to 50m.

Powered Pallet Truck

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

COUNTER BALANCED TRUCK

Counter Balanced Truck



Characteristics

- Most widely used, general-purpose warehouse trucks.
- Used for travel above 50m.
- Average capacity 1.5t to 10t.
- Can lift to average 7m.
- Operate in aisle width of 4m.

Forklift

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

STACKER

Stacker	Characteristics
	• Can lift up to 5m • Capacity range from 1t to 2t • Various configurations – walking, ride-on

Stacker

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

REACH TRUCK

Reach Truck	Characteristics
	• Used to reduce aisle space and use cube. • Typically operate in aisles 2.5m. • Can lift up to 8m.

Reach Truck

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

VNA TRUCK

VNA Truck  Intelligent as the VNA Truck is the warehouse technology to change and has been the wider model. Technology for work from 10 years, with over 100,000 units sold. High quality and low cost. (VNA Truck) is the most popular in the market.	Characteristics • Used to assemble less than pallet load quantities. • Items picked out of bins or racks in narrow aisles. • Typical lifting height 5m or more. • Capacities up to 1t. • High volume, high lift operations may use guide rails or guide wire or rails.
---	--

VNA Truck

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

PALLET TYPES & SPECIFICATIONS

PALLET STYLES



Dimensions, mm(W x L)	EURO Pallet type
800 × 1,200	EUR, EUR 1
1,200 × 1,000	EUR 2
1,000 × 1,200	EUR 3
800 × 600	EUR 6

Dimensions, mm (W x L)	Region Most Used
1219 x 1016	North America
1000 x 1200	Europe, Asia
1140 x 1140	Australia
1067 x 1067	North America, Europe, Asia
1100 x 1100	Asia
800 x 600	Europe (fits many doorways)

CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

CONVEYORS



CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

PALLETIZER / SHRINK-WRAP MACHINES



Palletizer



Shrink Warp Machine



CHAPTER 3: WAREHOUSE LAYOUT, STORAGE, HANDLING & MHE

CASE STUDY 01: HOW TO CALCULATE COST IN VAS AND MAKE QUOTATION

You are Business Development in DC for FMCG. Currently, Your customer is requested VAS – packaging by combine in 1 combo (1 bottle of 800 ml shampoo + 1 bottle of 200 ml). Forecast quantity is 90 000 package.

The information as below:

- Timeline process is requested complete in 7 days.
- The working capacity of warehouse is 250 products / person / day
- Worker's wage is 20 USD / person / day.
- 20 products packed into 1 carton, carton cost is 0.5 USD / Carton
- 1 roll of adhesive tape is 10 barrels, the tape cost is USD 0.25 / roll
- Electricity and water costs are 500 USD / month
- Management fee is 1,000 USD / month

- Please calculate the cost to complete this order on time
- To quote for this service, expected profit is 15%

